

Yuli Multi Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

RWTHAACHEN
UNIVERSITY



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Reverse Engineering

Data Generation

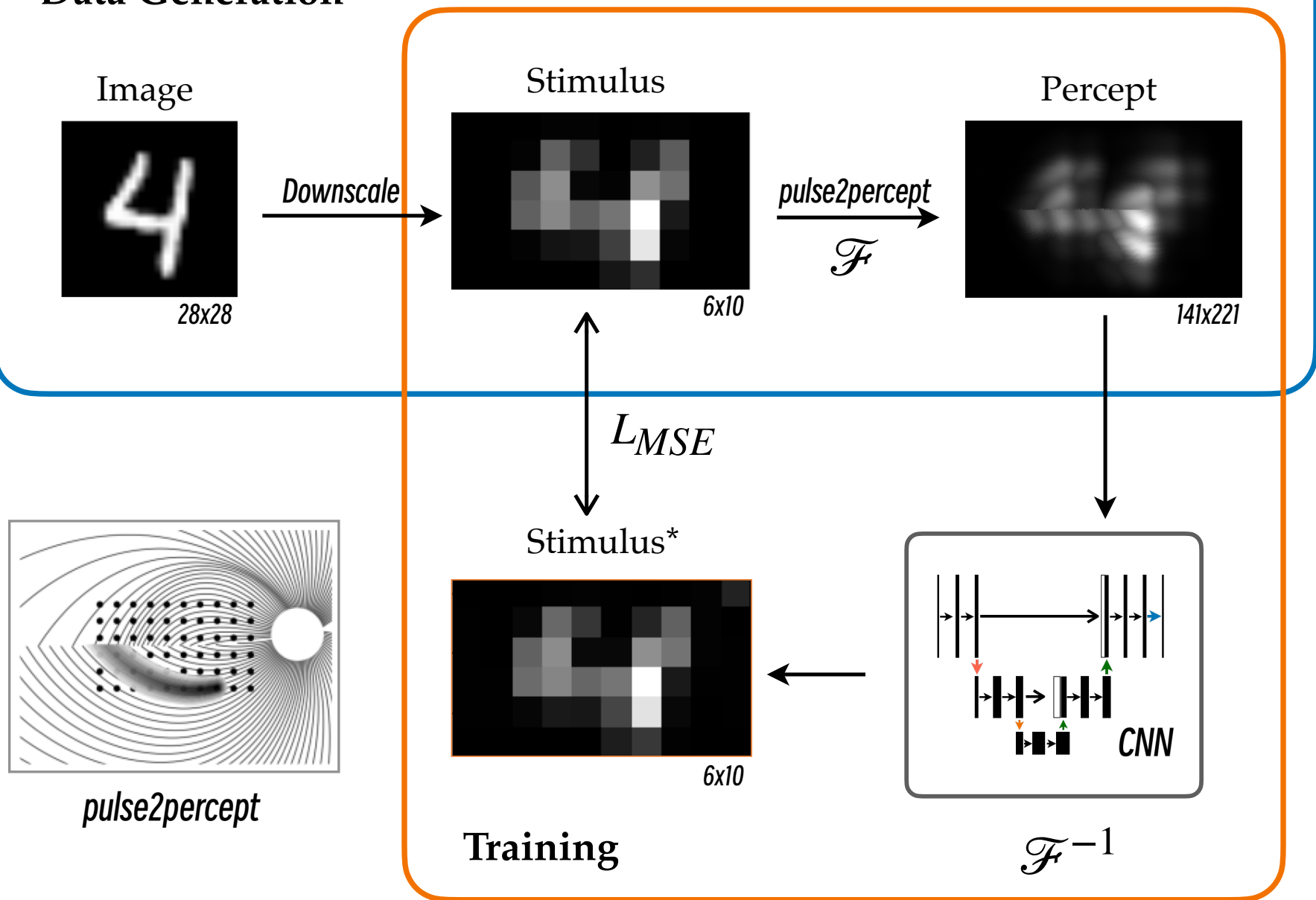
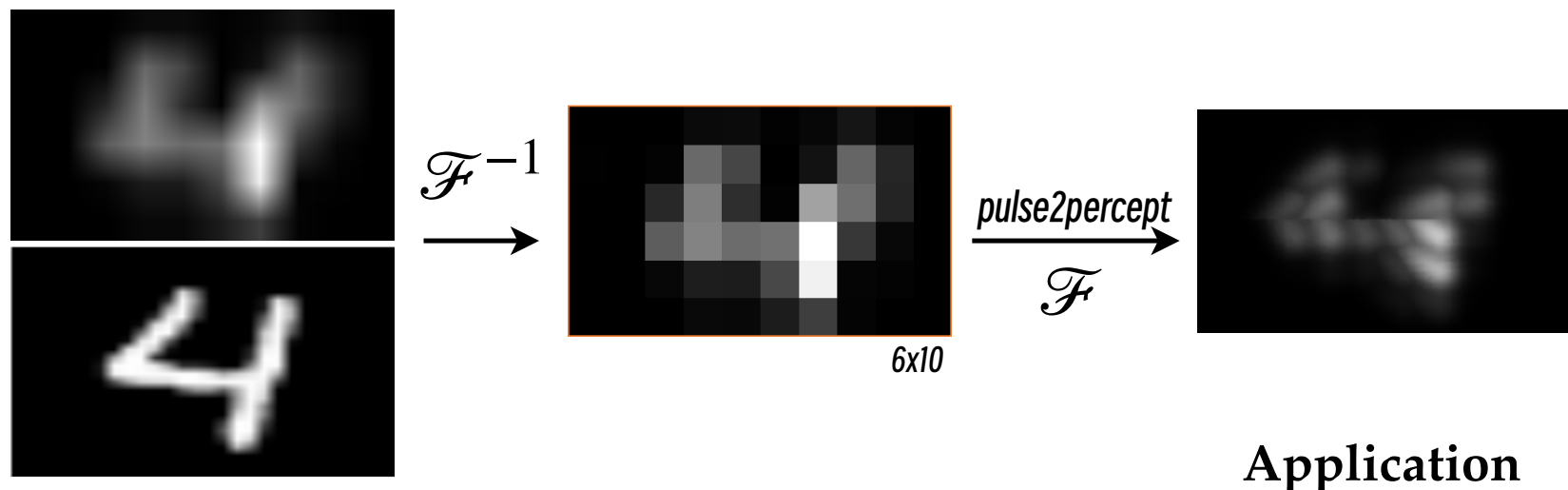
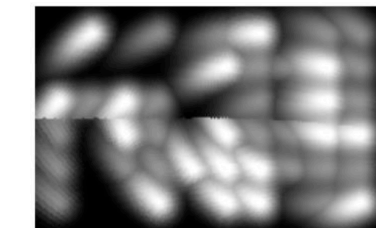
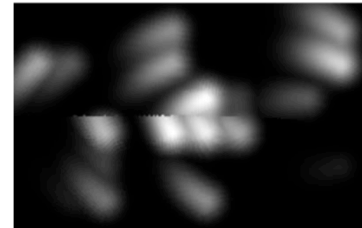
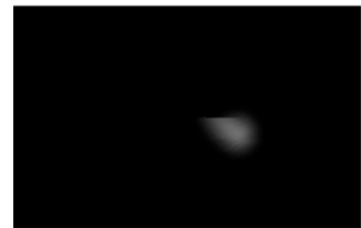


Image / Percept*



Application



Single electrode stimulation
with continuous values

Multi electrode stimulation
with continuous values

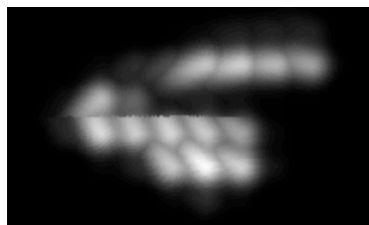
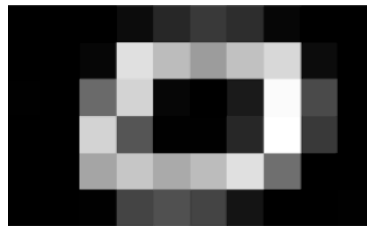
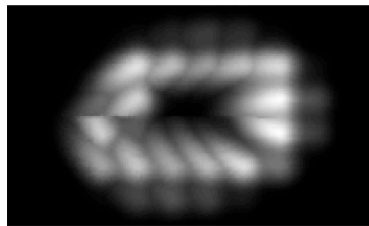
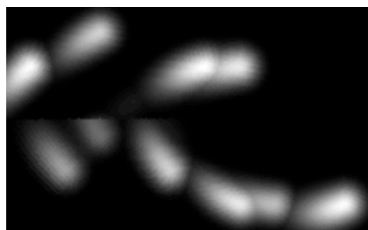
Multi electrode stimulation
with ternary values

Training

Percept

GT Stimuli

Predicted Stimuli

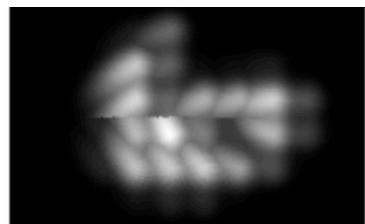
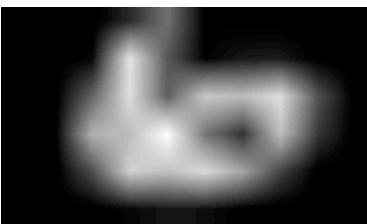
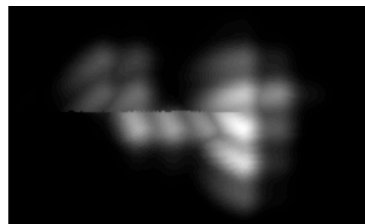
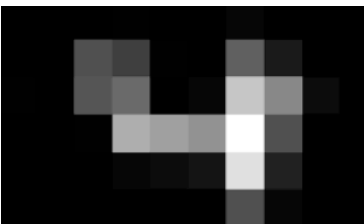
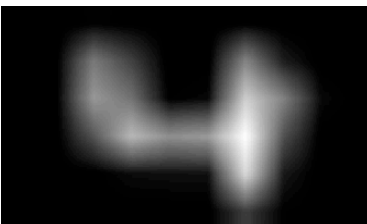
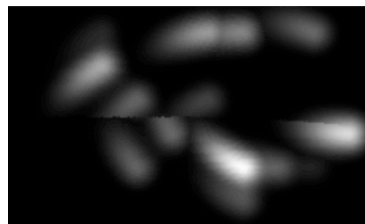
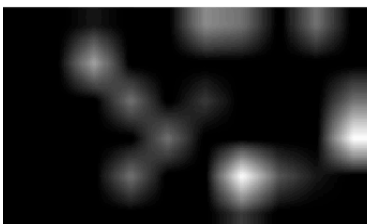
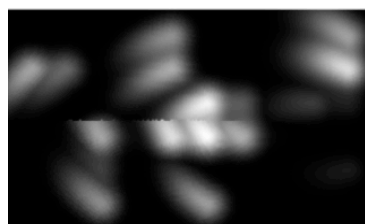
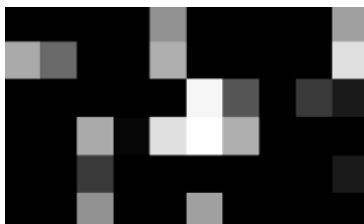
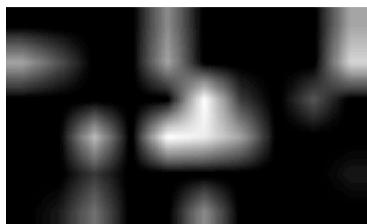


Application

Input

Stimuli

Percept



[Wuetai., 2023a]

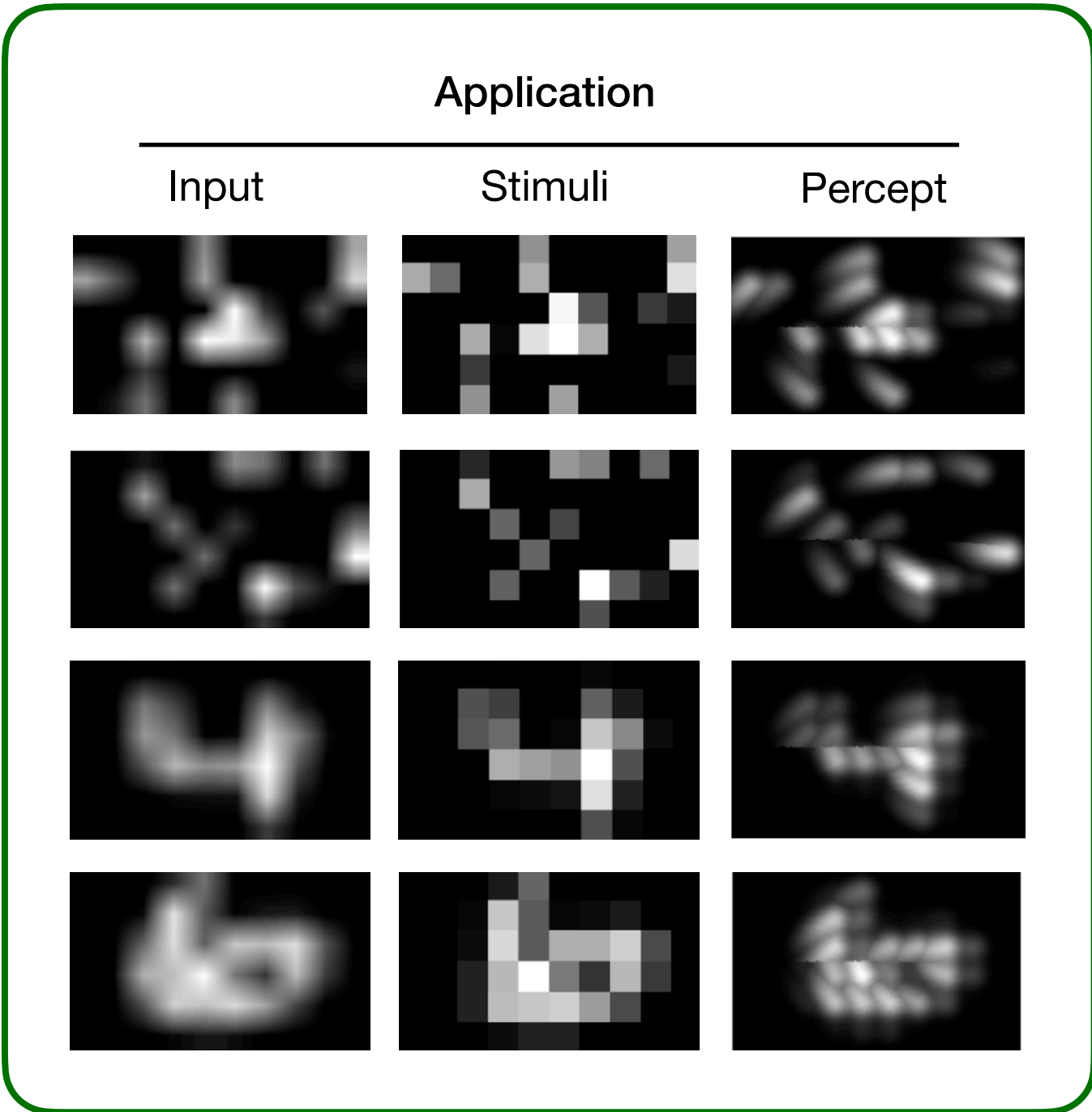
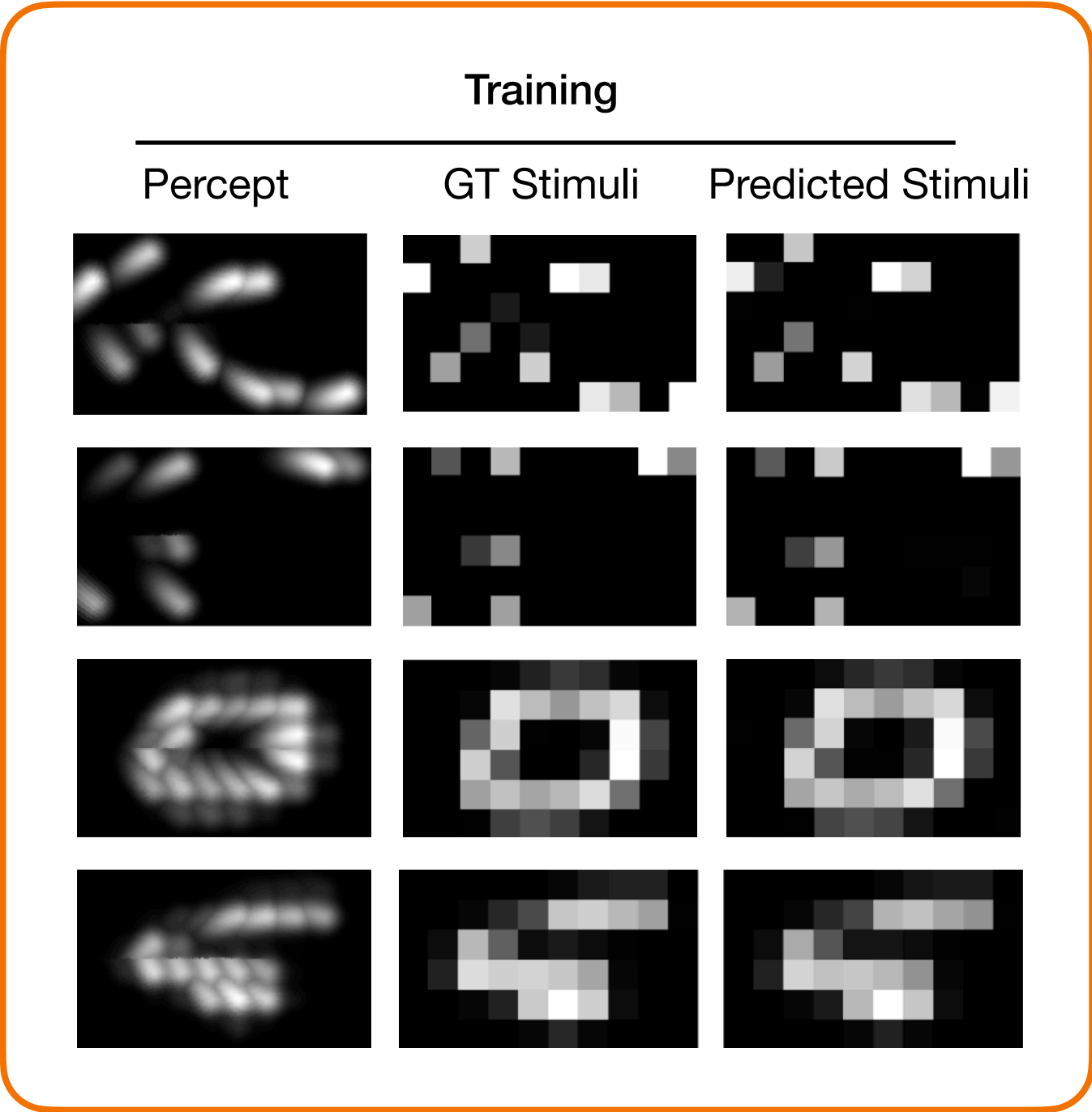
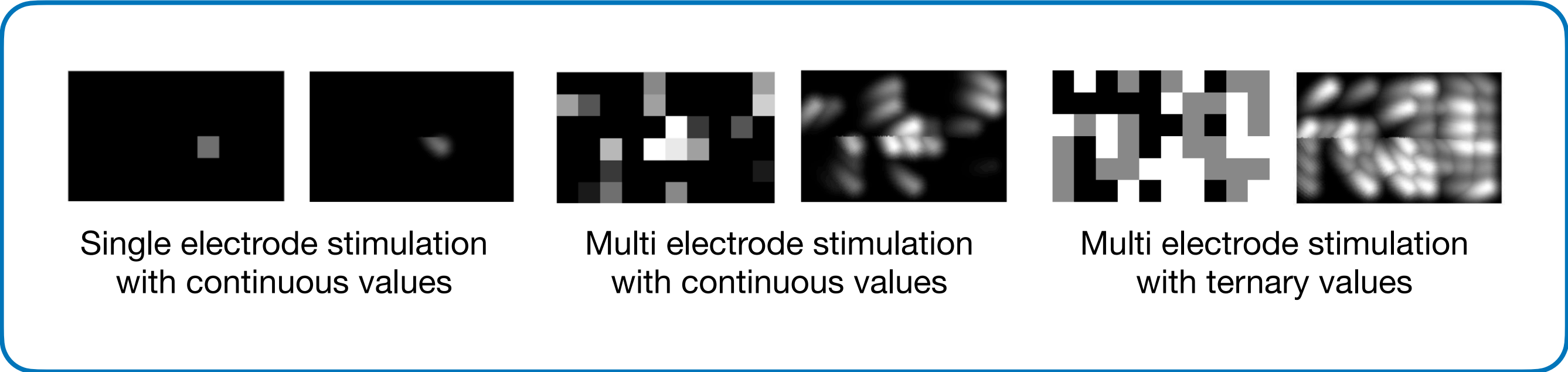
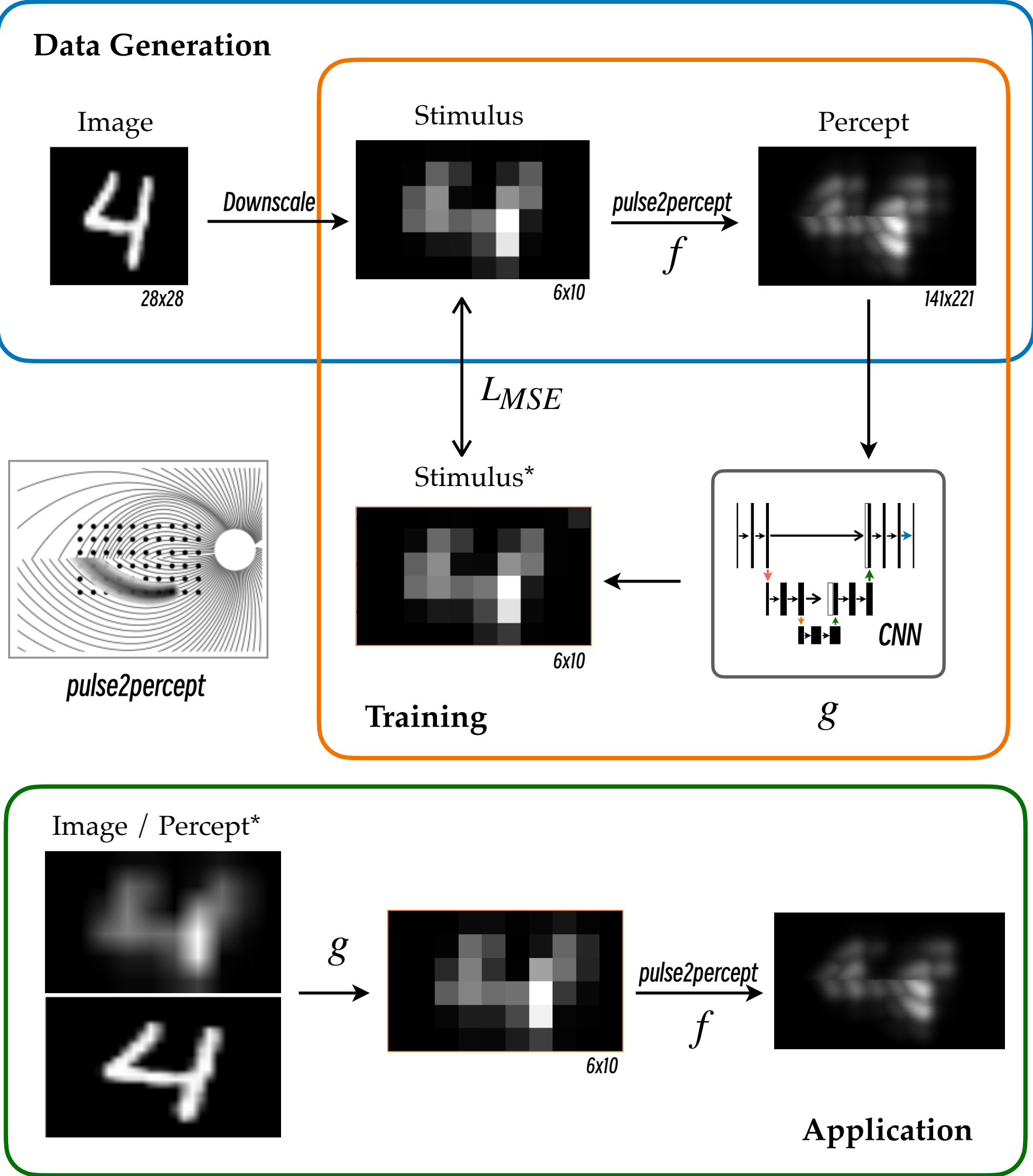






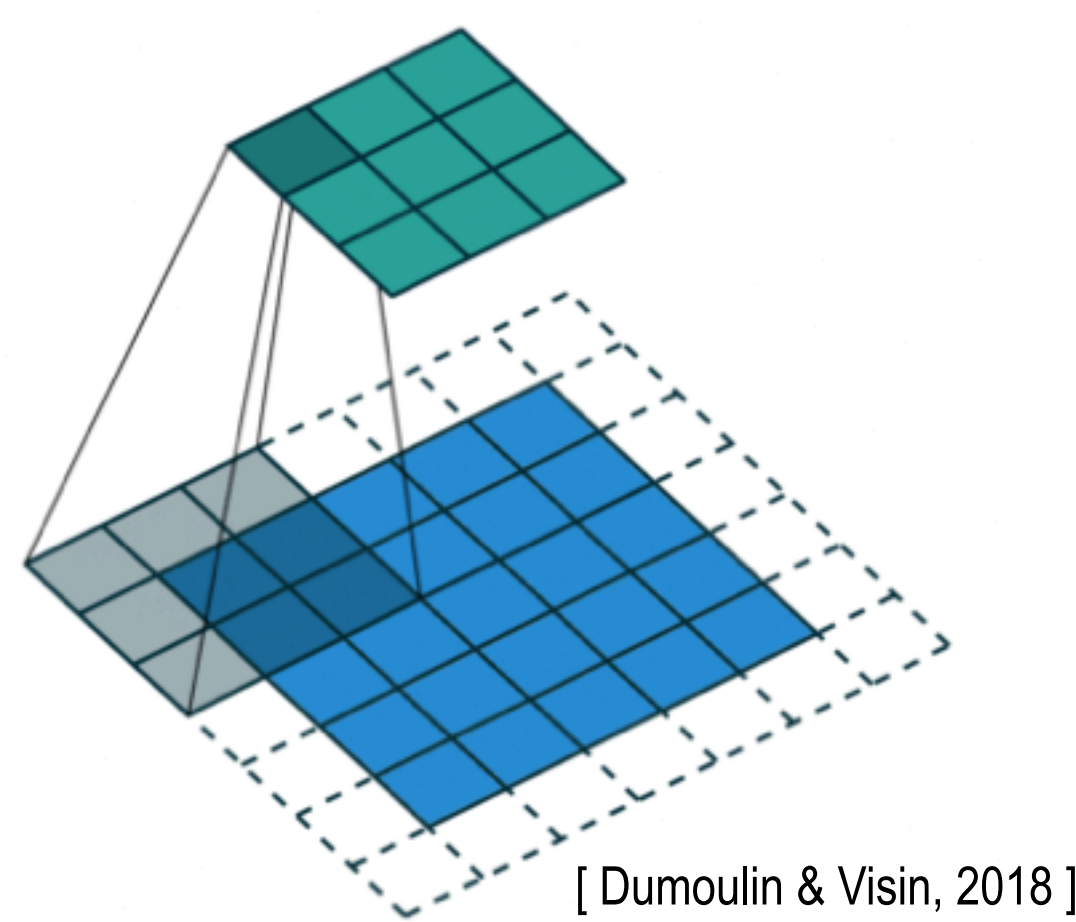
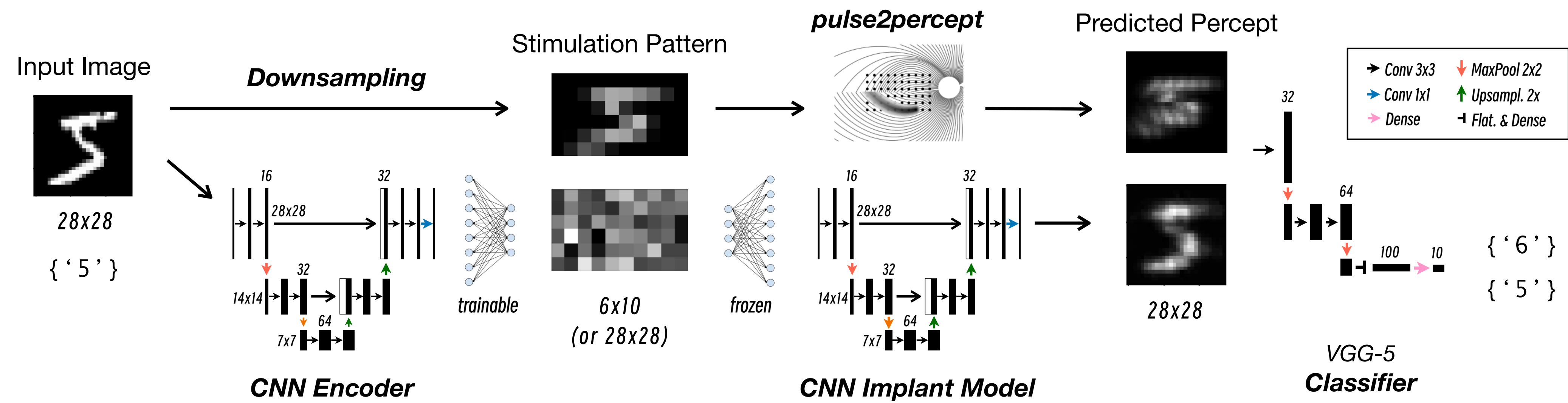


Reverse Encoding

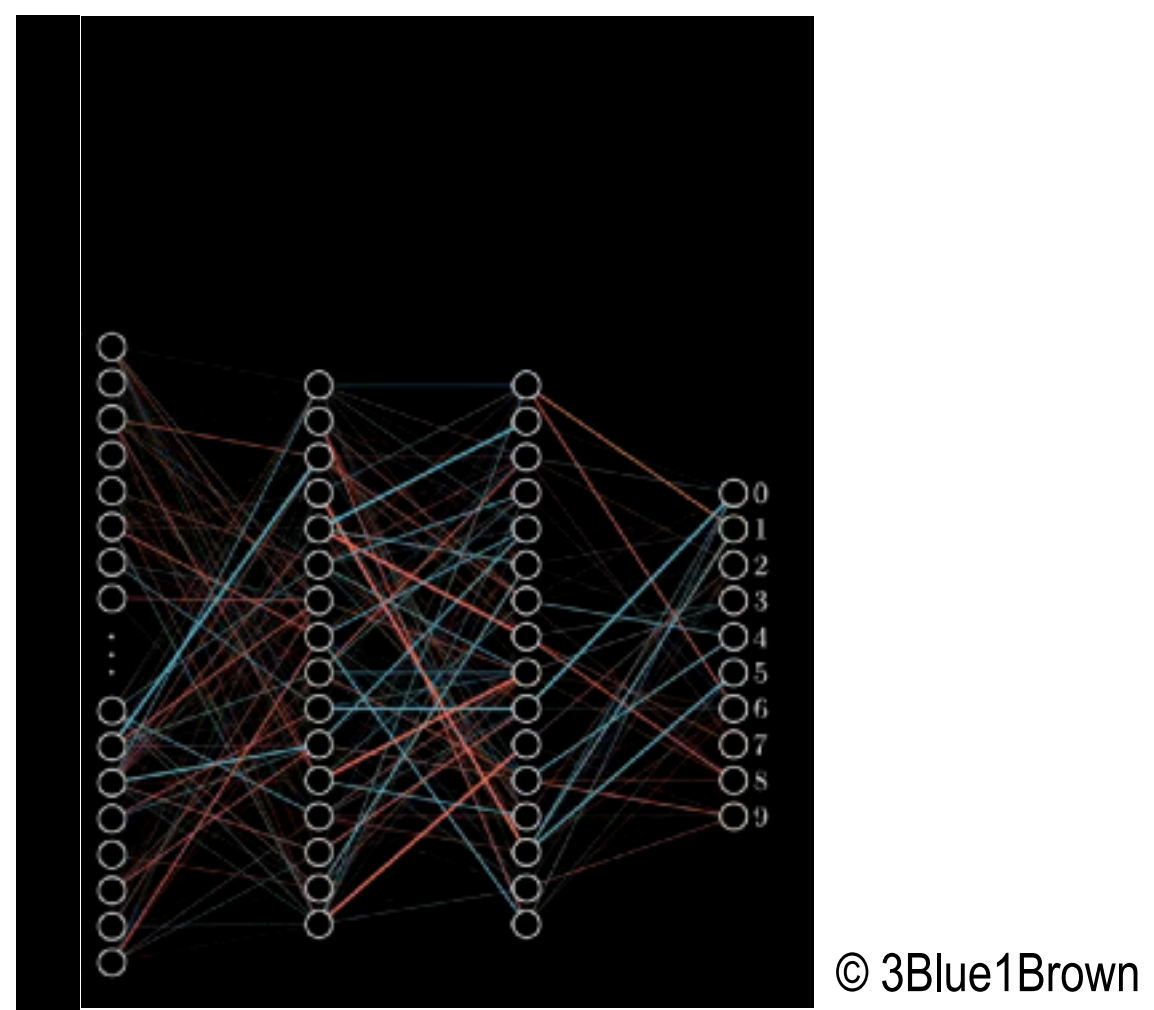


[Wu et al., 2023a]

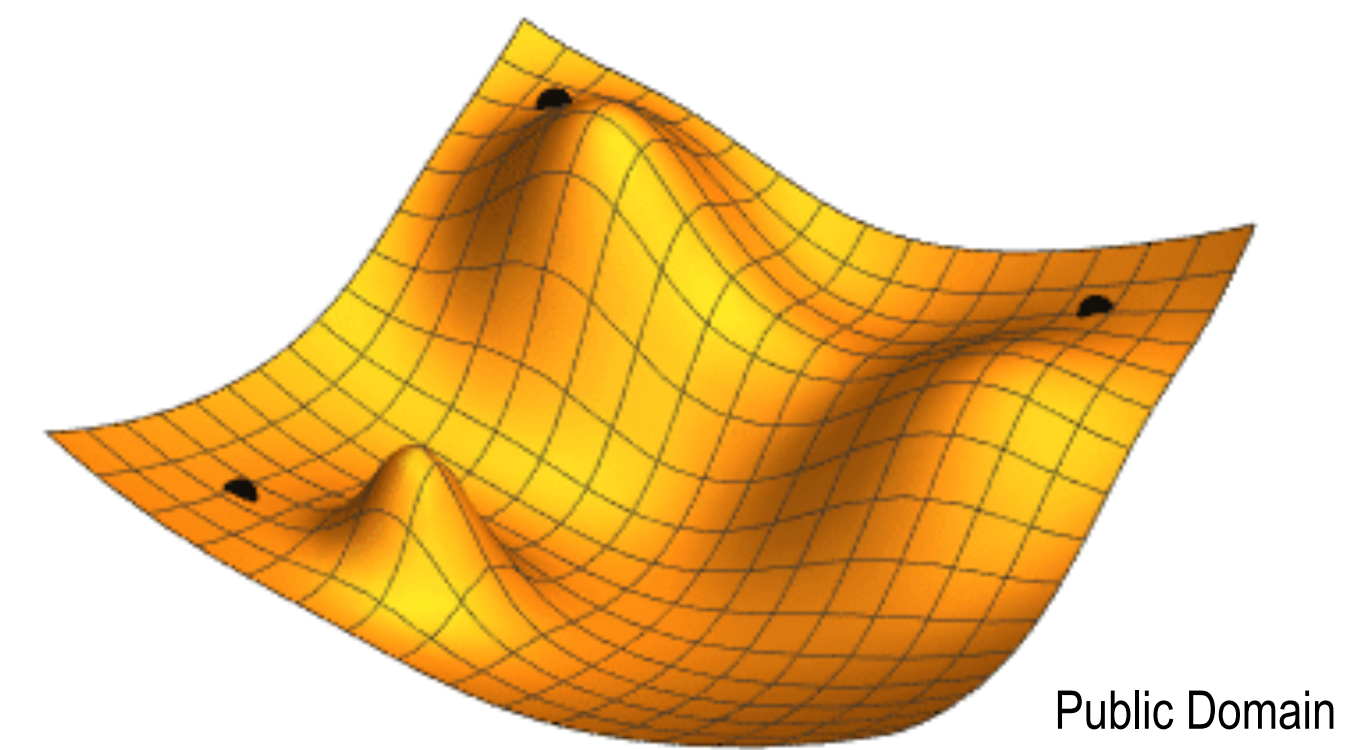
End-to-End Encoding



[Wu et al., 2023b] **Convolutional Layer**



Model Updating



Gradient Descent